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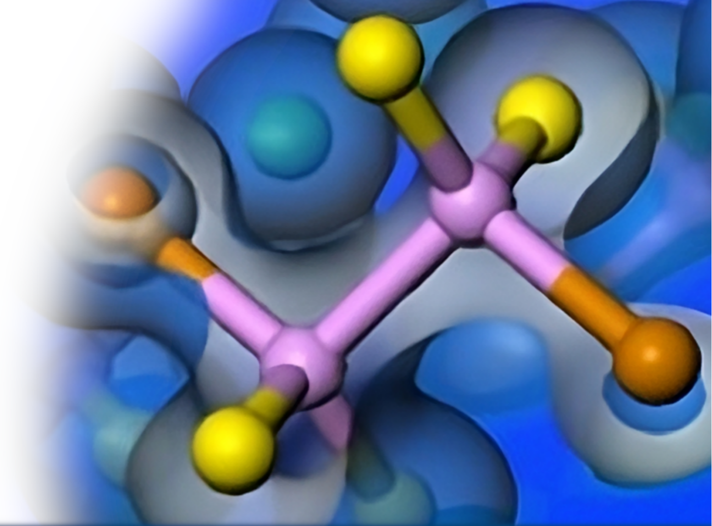

Phonons and vibrational spectroscopy

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What are experiments?

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Phonons
FD
DFPT
Summary

Energy and momentum

A wave with frequency ω and wavelength λ has energy E and momentum **p** of:

$$\begin{aligned} E &= \hbar\omega \\ \mathbf{p} &= \hbar\mathbf{q} \text{ where } |\mathbf{q}| = \frac{2\pi}{\lambda} \end{aligned}$$

What might be typical values?

Photon

$\omega \sim 400 - 800 \text{ THz (visible light)}$
$\lambda \sim 400 - 800 \text{ nm (visible light)}$
E is large, λ is large (if you're an atom!) so **q** is small,
$\longrightarrow$ light has a lot of energy, but not much momentum

Energy and momentum

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What might be typical values?

Phonon

$\omega \sim 0 - 10$ THz
$\lambda \sim 0.5$ nm (lattice vector)
E is small, λ is small so $\mathbf{q}$ is large,
$\longrightarrow$ phonons don't have much energy but do have lots of momentum.

Springtime

If an atom is pushed off its equilibrium position by a displacement **u**, it experiences a force pulling it back.

For a small displacement, we can model this as a spring, where the force is proportional to the displacement:

$$\mathbf{F} = -k\mathbf{u}$$

The energy of this displacement goes up as displacement squared:

$$E = \frac{1}{2}\mathbf{u} \cdot k\mathbf{u}$$

Force constant matrix

In general, all the atoms will be displaced so **u** is a vector of length $3N$.

The "spring constant" doesn't have to be the same in all directions, or for all atoms – it becomes a *matrix* called the force constant matrix (FCM).

$$E = \frac{1}{2}\mathbf{u} \cdot \mathrm{K}\mathbf{u}$$

where K is the force constant matrix. It is always symmetric, i.e. $\mathrm{K}_{ij} = \mathrm{K}_{ji}$.

If the matrix K is diagonal, then each atom moves independently, as if it only had its own spring to its equilibrium position.

In real life, atoms interact! There are also “springs” between the atoms, which means K has off-diagonal elements as well. What does this mean?

The atoms no longer move *independently*, their motion is coupled... there *are* still ways to move the atoms which are independent of each other, but these involve moving all the atoms together in particular ways.

They are collective modes – phonons

Independent atom example

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Phonons

Recall that if the matrix K is diagonal, then each atom moved independently.

The phonon modes are the collective modes which diagonalise K, i.e. they are the eigenvectors of the force constant matrix.

Their eigenvalues are the mass times frequency squared of that phonon mode.

Thus to calculate the phonons from K, we need to solve

$$\mathrm{K}\mathbf{u} = m\omega^2\mathbf{u},$$

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and when we say "we" we mean "a computer"!

Collective motion example

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3D Phonon modes

For N atoms, there are $3N$ ways we can move them, which leads to $3N$ phonon modes.

Modes can be:

- **Longitudinal (L)**
  The displacement of the atoms is parallel to the direction of the wave, $\mathbf{q}$
- **Transverse (T)**
  The displacement of the atoms is perpendicular to the direction of the wave, $\mathbf{q}$

3D Phonon modes

For N atoms, there are $3N$ ways we can move them, which leads to $3N$ phonon modes.

- 3 **Acoustic modes (A)**
  All the atomic displacements are in the same direction. There is one longitudinal acoustic (LA) mode and two transverse acoustic (TA) modes.
- 3N-3 **Optical modes (O)**
  Not all the atomic displacements are in the same direction. There are longitudinal optical (LO) mode and transverse optical (TO) modes.

Why do we care?

Phonons tell us about the natural vibrations of our system. They are important for:

- Is our material mechanically stable?
- Thermodynamics
  - Free energy (vibrational entropy)
  - Specific heat capacity
- IR and Raman spectra
- Inelastic neutron scattering (INS)
- Electron-phonon coupling

Real phonons: α-quartz

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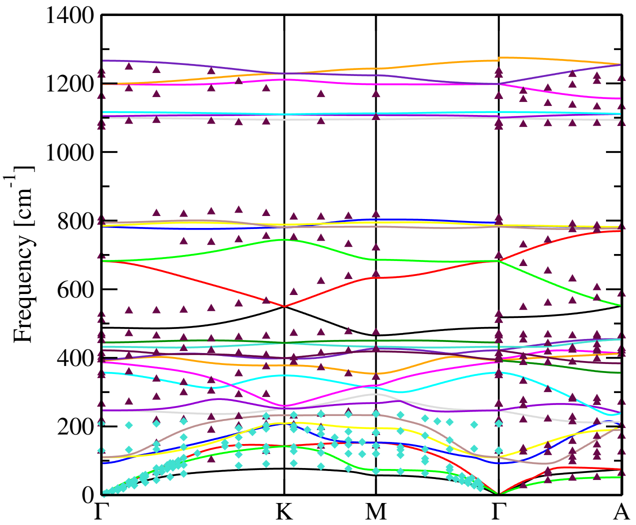
- Phonons
- FD
- DFPT
- Summary



Force constant matrix

How do we know what the force constant matrix should be for a particular material?

Force constant matrix

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$$
\begin{aligned}
E &= \frac{1}{2}\mathbf{u}\cdot\mathrm{K}\mathbf{u} \\
&= \frac{1}{2}\sum_{ij}\mathrm{K}_{ij}u_i u_j \\
\Rightarrow \frac{\partial E}{\partial u_k} &= \frac{1}{2}\sum_{i}\mathrm{K}_{ik}u_i \\
\Rightarrow \frac{\partial^2 E}{\partial u_m \partial u_k} &= \mathrm{K}_{mk}
\end{aligned}
$$

so K is the matrix of second derivatives of the energy, with respect to moving atoms.

Force constant matrix

We can also write it in terms of the force **F** on an atom,

$$\begin{aligned} F_k &= -\frac{\partial E}{\partial u_k} \\ \Rightarrow \mathrm{K}_{mk} &= \frac{\partial^2 E}{\partial u_m \partial u_k} \\ &= \frac{\partial}{\partial u_m}\frac{\partial E}{\partial u_k} \\ &= -\frac{\partial}{\partial u_m} F_k \end{aligned}$$

In other words...

K_{mk} tells us how the force on atom k varies as we move atom m.

Since the matrix is symmetric, it also tells us how the force on atom m varies as we move atom k.

Finite displacement

Method 1:

- Optimise geometry so all forces are zero
- Displace atom m a little bit
- Compute the new forces
- Since $K_{mk} = -F_k$, the forces tell us the whole mth row of the matrix
- Repeat for a different atom displacement
- There are $3N$ displacements in total (x, y and z for each atom)

(We actually use a more accurate version: we displace each atom forward and then backwards, so there are $6N$ displacements in total.)

Finite displacement

- Works with all pseudopotentials
- Works with all XC functionals
- Naïve way only gives modes with $\mathbf{q} = 0$
  Need to construct a supercell to get other $\mathbf{q}$, but CASTEP does this for you

In your param file:

```
task: phonon
phonon_method: finitedisplacement
phonon_fine_method: interpolation
```

The Force Constant Awakens

How do we know what the force constant matrix should be for a particular material?

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The Force Constant Awakens

How do we know what the force constant matrix should be for a particular material?

$$\mathrm{K}_{mk} = \frac{\partial^2 E}{\partial u_m \partial u_k}$$

We can compute the derivatives of E using perturbation theory.

In DFT this is complicated because as we move an atom the density changes, which also changes the Hamiltonian.

We use **Density Functional Perturbation Theory (DFPT)**.

[This maths intentionally left blank]

Dynamical matrix

It's more convenient to compute the Fourier transform of the force constant matrix - called the dynamical matrix.

This isn't just mathematically convenient, it also lets us compute the phonon modes for any wavevector $\mathbf{q}$.

There are still $3N$ ways to move the atoms, so there are $3N$ perturbations,

Phonons
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DFPT
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Method 2:

- Optimise geometry so all forces are zero
- Consider perturbation where we displace atom m a little bit
- Use DFPT to calculate K_{mk} from the energy derivatives
- Repeat for a different atom displacement
- There are $3N$ displacements in total (x, y and z for each atom)

DFPT

- Only implemented for norm-conserving pseudopotentials
- Only implemented for semi-local XC functionals
- Gives all modes for all **q**

In your param file:

```
task: phonon
phonon_method: dfpt
phonon_fine_method: interpolation
```

q-points

How do you tell CASTEP what **q**-points to use? Similar way to the DFT DOS and bandstructure:

In the cell file, e.g. to get a $4 \times 4 \times 4$ grid of **q**-points for phonon DOS:

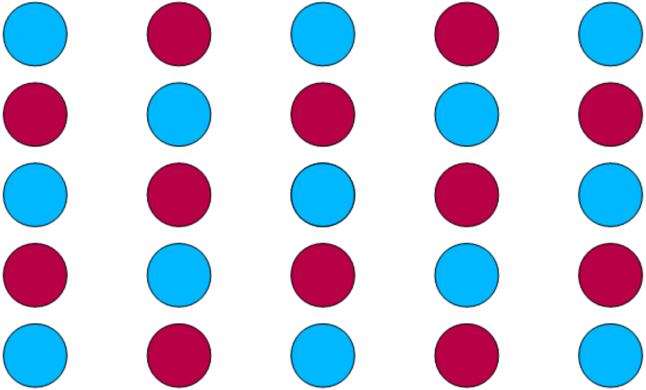
```
phonon_kpoint_mp_grid 4 4 4
phonon_fine_kpoint_mp_grid 6 6 6
```

("fine" grid is for interpolation) or for a phonon dispersion ('bandstructure')

```
%block phonon_kpoint_path
[list of high-symmetry points]
%endblock phonon_kpoint_path
```

Note that the keyword uses 'kpoint' rather than 'qpoint' (sorry!)

When a vibration travels through a crystal, it creates a periodic distortion to the atoms.



Phonons

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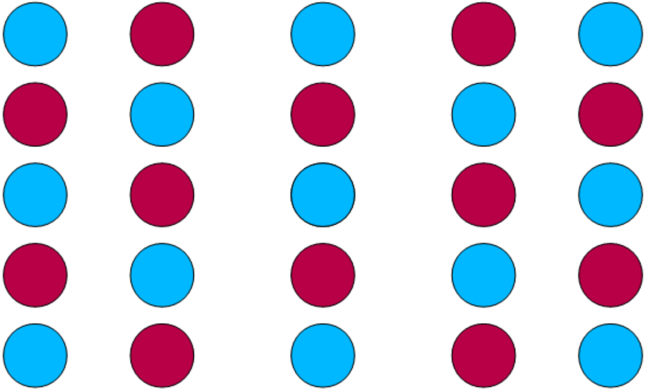

Phonons

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Periodic distortion $\longrightarrow$ wavevector **q**.
Periodicity not required to be the same as the unit cell.

Atoms are displaced from equilibrium positions by the vibration. The displacement *d* is real, not complex, so:

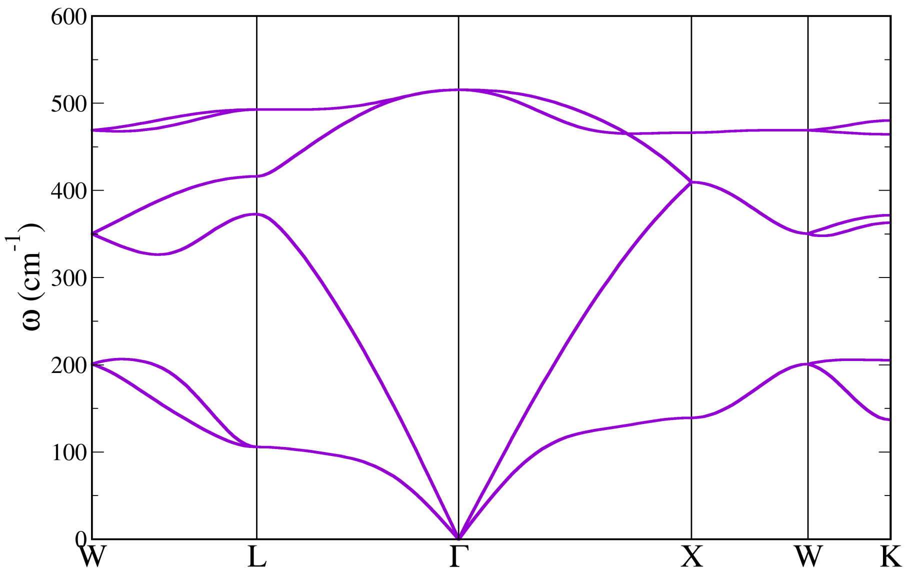
$$d_q(\mathbf{r}) = a_q \cos(\mathbf{q}.\mathbf{r})$$

The natural vibrations of a material are called *phonons*.

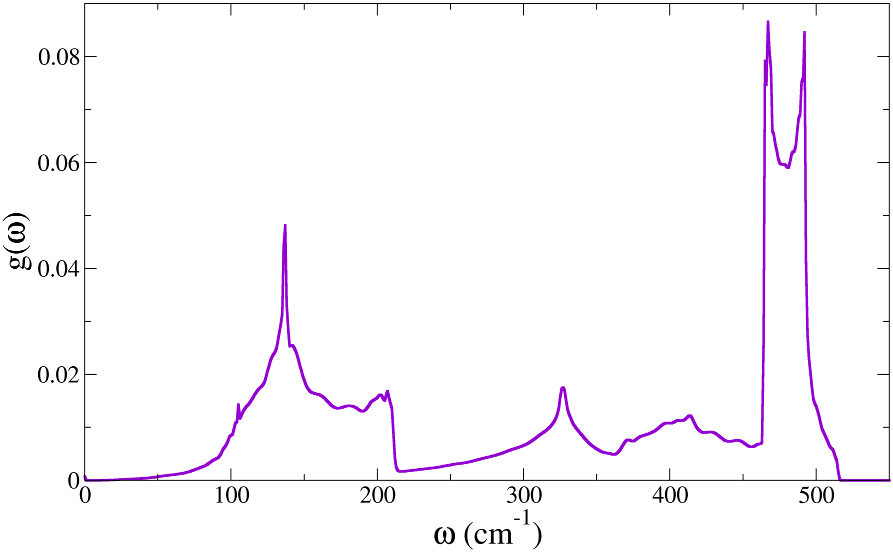
Phonon energies depend on **q**: plot *phonon* band structure (usually as frequency $\omega(\mathbf{q})$ against **q**, where $E = \hbar\omega$).

Shows frequency of different lattice vibrations, from long-wavelength acoustic modes to the shorter optical ones.

Si phonon bandstructure



Si phonon DOS



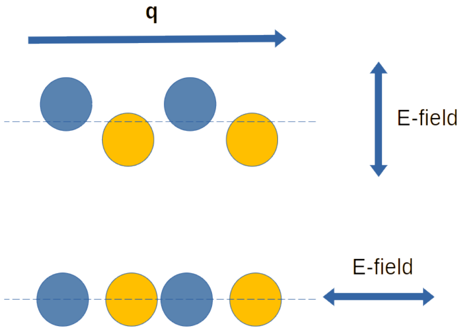
LO-TO splitting

There is one bit of physics which can cause some trouble.

If your materials has charged ions, when you move one it creates an electric dipole.

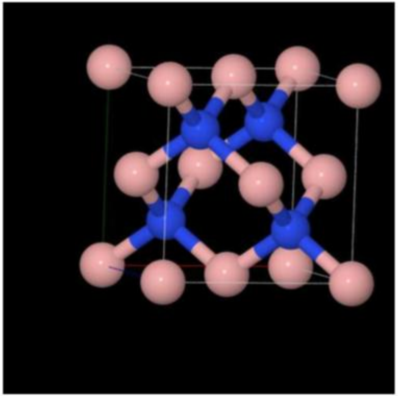
This electric field creates an extra force, but it depends on the *direction* of the displacements.

LO-TO splitting

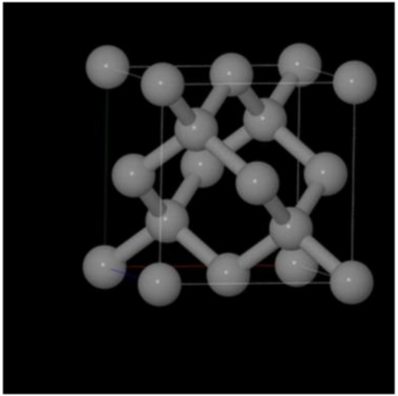


- Different effect depending on E-field ϕ direction relative to $\mathbf{q}$
- Transverse (TO), $\phi \cdot \mathbf{q} = 0$ and no effect
- Longitudinal (LO), $\phi \cdot \mathbf{q} \neq 0$ and extra restoring force
- Depends on the dielectric permittivity

- Two similar structures



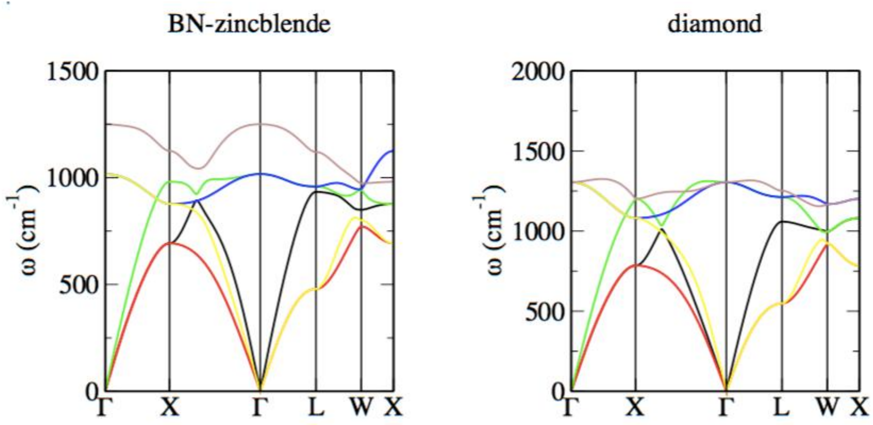
Zincblende BN



Diamond

LO-TO splitting

- Note degeneracy of mode at Γ



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Need to know the Efield response of your material, so in your param file:

```
task: phonon+efield
```

- Computes the dielectric permittivity
- Also gives the lattice contribution to the $\omega \to 0$ response
- Get an extra output file (`.efield`) for IR part

LO-TO splitting NaCl

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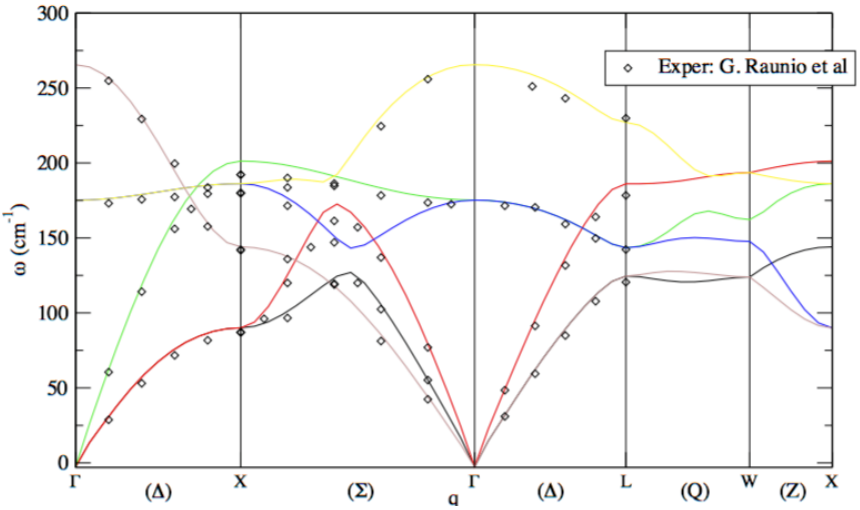
Phonons

FD

DFPT

Summary



What to watch out for

Phonon calculations are particularly sensitive to numerical noise, so your calculations need to be especially good. In particular:

- Good **k**-point sampling for the DFT
- Large “fine grid” for the potentials, eg in your param file
  `fine_grid_scale: 4.0`
- For DFPT, need very well-converged DFT, eg in param file
  `elec_energy_tol: 1.0e-9 eV`
- Make sure the geometry is well converged

We’ve considered the second derivative of the energy, “harmonic phonons”. Sometimes higher-order (“anharmonic phonons”) effects are important.

Summary

Phonon dispersion (band structures) show how phonon frequencies (proportional to energy) change across the 3D Brillouin zone (ω against $\mathbf{q}$)

Phonon DOS: sample the *entire* Brillouin zone

Need well-converged calculations.

Can predict a large number of experimental spectra, but beware LO-TO splitting!